

On the Functional Characterisation of Polymer Electrolyte Fuel Cells: Part II. Impact of Air Uncertainty on Localized Temperature Distributions

Amit M. Manthanwar^{*a,d}, Thiago Lopes^b, Stephen C. Atkins^c,
Anthony R. Kucernak^c, Efstratios N. Pistikopoulos^d

^a*Department of Chemical Engineering, Imperial College London, United Kingdom*

^b*Institute of Electrochemistry, University of São Paulo, São Carlos, Brazil*

^c*Department of Chemistry, Imperial College London, United Kingdom*

^d*Energy Institute, Texas A&M University, College Station, Texas*

Revision 3: May 25, 2016

Abstract

In order to gain the unprecedented insights into the inner workings of a fuel cell, in our previous work we developed a novel in situ experimental technique with the highest possible spatiotemporal resolution to map the localized temperature evolution that is mainly due to reaction only and also other factors. Using the technique, we identified point-in-location as well as point-in-time intensity of hotspots and concluded that the temperature hotspots are one of the main reasons for fuel cell's accelerated degradation leading to its eventual failure. In this work, we employ this novel in situ experimental technique to empirically investigate the impact of operational uncertainty arising due to variation in air stoichiometry and its relative humidity. Taking prognostic view, this variation in the oxidant feed is one of the most logically possible sources of uncertainty in an operating fuel cell. Our results show that the variation in cathode air supply has significantly stronger impact on the intensity of hotspots. Using these results we are able to qualitatively as well as quantitatively identify the optimal operating regime on traditional polarization curves augmented by temperature axis. This information is useful in material synthesis to avoid degradation as well as to improve cell's performance and lifetime. Understanding the impact of uncertainty is also essential while designing a robust feedback control strategy to mitigate its severe consequences.

keywords: fuel cell, electrochemistry, temperature measurement, heat management

1 Introduction

Fuel cell operates over a wide range of operating conditions where uncertainty is ubiquitous. The major source of uncertainty in operating fuel cell emanates from the variation in inlet air stoichiometry and its relative humidity. Air stoichiometry between maximum to nominal and relative humidity from ambient to saturated is of utmost importance as it has detrimental effect on steady state and dynamic performance. Study of fuel cell operational dynamics is not only important to understand material degradation but also in designing optimal control strategies. Impact of uncertainty on the transient behaviour of fuel cell is a widely researched topic both in experimental and theoretical modelling works. There exist numerous studies on the transient

^{*}The financial support from EPSRC grants (EP/I014640/1 and EP/K503381/1), sponsorships from CPSE industrial consortium and Texas A&M energy institute is gratefully acknowledged.

behaviour of fuel cell influenced by the effect of air stoichiometry. Amongst them, Kim *et al.*, [1], studied the experimental impact of normal, starved and excess air stoichiometry on current density. They modeled the empirical input output response in Laplace domain as first order and second order systems. They observed that air stoichiometry affects the time constant and magnitude of overshoot peaks. Yan *et al.*, [2] found out that the air stoichiometry influences performance directly by supplying oxygen and indirectly by influencing humidity which affects water flooding on cathode side. Their experiment showed that the optimum air stoichiometry was much larger than the stoichiometric value required for oxidation of the fuel. Using neutron imaging Iranzo *et al.*, [3], found that the cathode relative humidity had a much larger effect on the cell water content and overall performance than the anode relative humidity. Clearly, both low and high relative humidity are detrimental to fuel cell. At low relative humidity the operational performance decreases and high humidity causes accelerated material degradation. Bi *et al.*, [4], experimentally observed a loss rate of three times the cathode catalyst mass and two times the active surface area at 100% RH compared to 50% RH.

Using numerical simulations, Qu *et al.*, [5], showed that the dynamic undershoots or overshoots which are also observed experimentally by [1] can be eliminated by varying the air stoichiometry. In all these studies the ‘bulk measurement’ of temperature was used. In the recent national FCEV learning demonstration report Wipke *et al.*, [6], have reported that 40% of fuel cell failures are due to the thermal management issues. In our previous work, [7], we developed a novel in situ experimental technique to measure localized temperature distributions with the highest possible spatial and temporal resolution. We investigated the existence of hotspots and identified their point-in-space location and point-in-time intensity. We discovered that there exist distribution in reaction rate which gives rise to temperature gradients as the fuel and oxygen gets consumed during their journey from inlet to outlet with occurrences of localized peaks/hotspots in the region of fractional distance between 0.35 to 0.65 mm from the inlet. These hotspots under the impact of uncertainty may lead to formation of pinholes which causes irreversible damage to the cell by allowing fuel crossover and internal short circuit within MEA, [8]. Therefore, to make operating decision that are either ‘risk averse’ or ‘risk neutral’ or for some ‘risk seeking’ we must study the impact of uncertainty on the localized temperature distributions.

Recently Bodner *et al.*, [9], have studied the impact of pinhole locations. From our previous results, [7], we already know the locations of hotspots that may lead to pinhole formation. In this work, we utilize our novel experimental technique, [7], to study the experimental impact of most logically possible operational uncertainty in inlet air by varying its stoichiometry and relative humidity. The purpose of this study is to observe how the hotspots react to these uncertain variations which influences membrane humidity and cathode water flooding. This information is then used to identify the safe—risk averse, optimal—risk neutral and extreme—risk seeking regions for the fuel cell’s operation. From the system integration and automation point of view, this study is of particular importance in designing control strategy that can operate the fuel cell efficiently by immunizing the impact of uncertainty. From the material point of view, this study contributes to the fundamental understanding of failure mechanism due to thermal stresses.

The remainder of this paper is organized as follows: section 2 presents the details of experimental setup and test procedure; section 3 empirically investigates the impact of inlet air

uncertainty on the localized temperature distributions and also delves on the identification of optimal operating regions on experimentally obtained temperature dependent characteristic performance curves; finally section 4 concludes the paper with remarks on future research directions.

2 Experimental

The objective of this work is to study the transient response of a polymer electrolyte membrane fuel cell when the feed air stoichiometry is varied from ‘excess’ to ‘normal’ to ‘critical’. The normal case is when stoichiometry is 1.5 whereas excess is 2.5 and above. The critical case is when air stoichiometry is 1.0 that is the exact required amount, below which oxidant starvation occurs. Starvation causes permanent damage to the cell due to the loss of electrochemical surface area of the cathode catalyst, [10]. From the operations perspective, when feedback control is in place, we expect that the starvation condition is a situation, if occurred then control system would trigger an auto shut down sequence. Therefore, the focus of this work is on a detailed experimental study of uncertainty in air supply between realistic operating bounds so as to mitigate its impact by deploying robust control strategy for effective heat management.

The second objective of the experiment is to study the impact of air relative humidity varied between ambient condition of 50% to the saturated level of 100%. During the experiment, load was applied at constant voltage by a voltage step in the form of a staircase of 0.8 to 0.4 V with 0.1 V interval as well as a rectangular wave between 0.8 to 0.5 V. Temperature response was measured both on anode and cathode side for stoichiometric variations. While for the relative humidity, response was recorded on cathode side to study water flooding phenomena in relation to temperature changes. Throughout the experiment hydrogen fuel stoichiometry, its relative humidity and uniform thermal bath was maintained constant at their respective values of 1.5 λ_{H_2} , 100% RH and 80 °C. During these step changes sufficient time was allowed for response to reach steady states. The rest of the procedure using our fully automated, highly instrumented experimental test platform is same as described in our previous work, [11].

In the first set of observations, fuel cell’s steady state performance characteristics were plotted as polarization curves with augmented temperature axis. These polarization curves were obtained by stepping the current density in forward as well as reverse direction, allowing the fuel cell to stabilize, and measuring the fuel cell voltage. In the second set of observations, transient response was recorded by, step changing the voltage, measuring the time response of current density and localized temperature distributions.

3 Results and Discussion

3.1 Fuel cell performance characteristics

Figure 1 presents the characteristic polarization curves where the uncertain region of operation is shown by the shaded areas. We observe a peak power density of 641 mW/cm² occurring at 0.43 V for 100% RH while for 50% RH the value is 572 mW/cm² at 0.51 V for constant air stoichiometry of 2.5. When the air stoichiometry is reduced to 1.0 at 50% RH the peak occurs at

425 mW/cm² at 0.5 V. For constant 0.5 V the power density values are 635.6 and 424 mW/cm² for 2.5/100% and 1.0/50% respectively, which means there is a 33% reduction in the power density value due to limited availability of oxygen and reduced proton conductivity in both the membrane and catalyst layer. We would need to compare 2.5/100% with 1.0/100% to discuss the differences in performance due to oxygen “starvation” only.

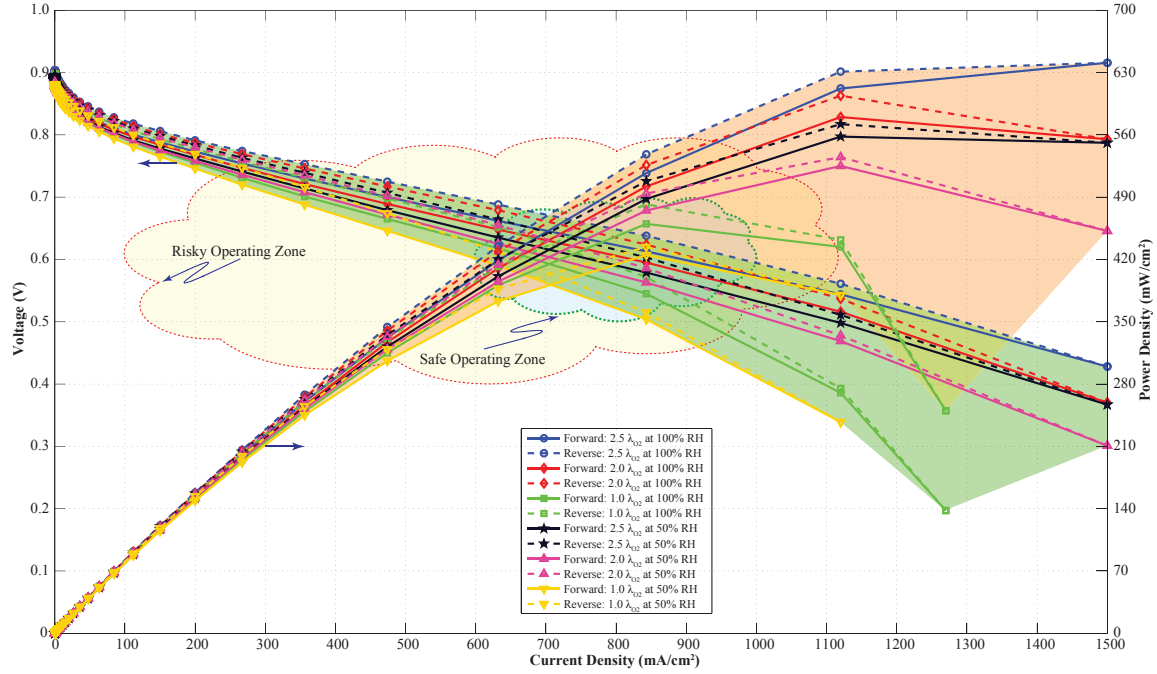


Figure 1: Performance characteristics for varying oxygen stoichiometry and relative humidity

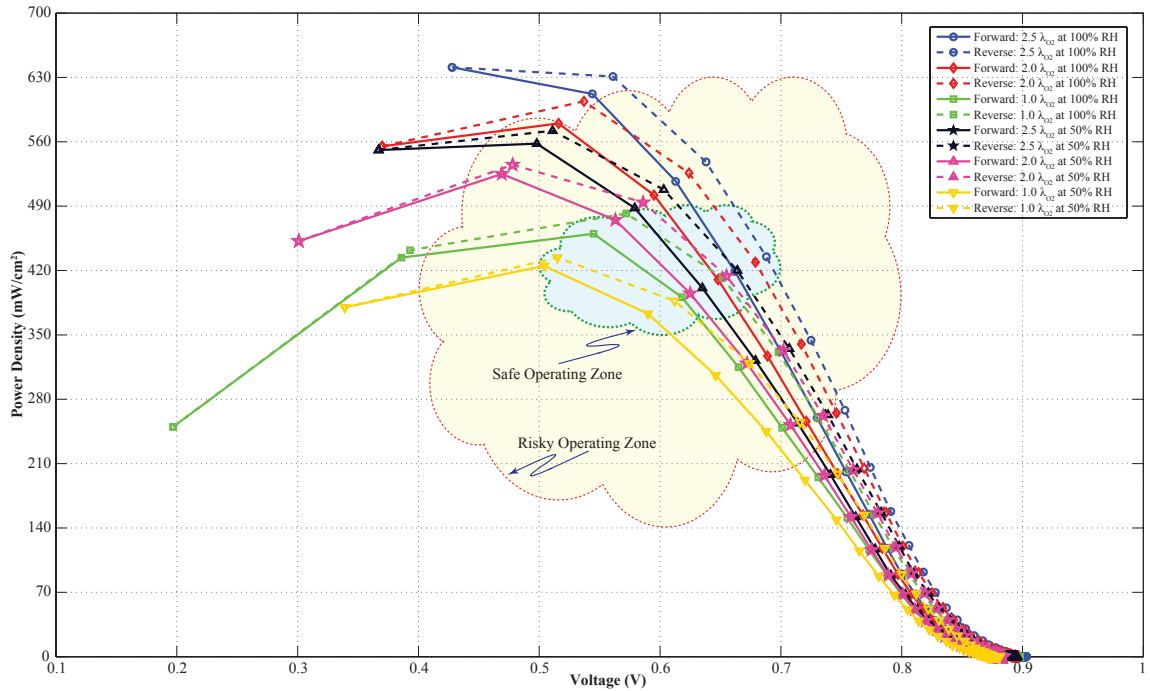


Figure 2: Power-voltage characteristics for varying oxygen stoichiometry and relative humidity

3.2 Effects of oxidant stoichiometry

At low oxidant stoichiometry the loss of performance is due to depletion of charge carriers since there is not enough oxygen present at cathode side for reaction to convert proton. Thus, mass transport dominates around 900 mA/cm^2 for low stoichiometry and low humidity. Figure 2 also helps us identify the approximate region of operation which are shown by the cloud of safe and risky operating zones. However, in order to find the exact optimal region of operation under the influence of uncertainty we must look at the temperature variations. These variations can be seen from the temperature dependent performance characteristics shown in Figure 3 as well as steady state and the dynamic temperature profiles presented in following section.

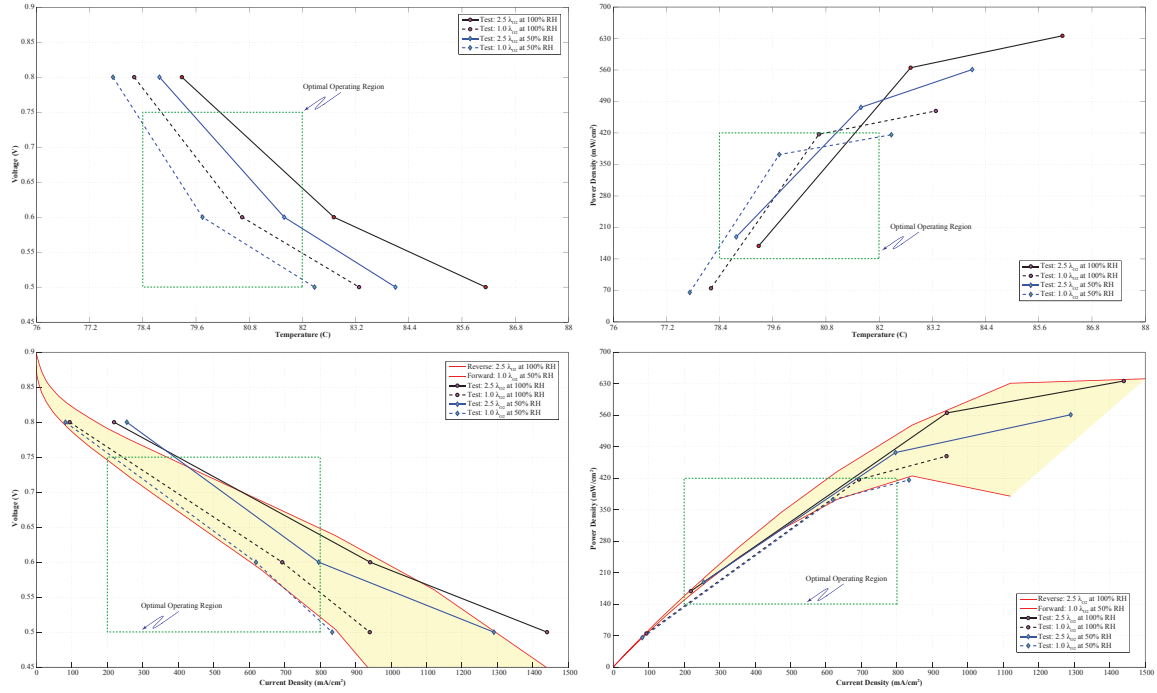


Figure 3: Temperature dependent performance characteristics under uncertainty

3.3 Impact of uncertainty on steady state temperature map

Figure 4 illustrates the steady state temperature distribution inside an operating fuel cell under the influence of uncertainty present in cathode air inlet. Although, the hotspots occurs at the same location, their intensity is affected by the air quality. These results reveals that at 0.6 V , regardless of uncertainty, the temperature appears to be almost uniform throughout and remains close to 80°C . These results clearly illustrates that the performance of the cell is best when operated within the safe operating zone of 0.5 to 0.7 V . The variation in temperature distribution along the fractional distance from the inlet is plotted in Figure 5 and corresponding difference in Figure 6. Notice that in Figure 5 the temperature peak shifts towards the inlet as the voltage reduces from 0.7 to 0.4 V .

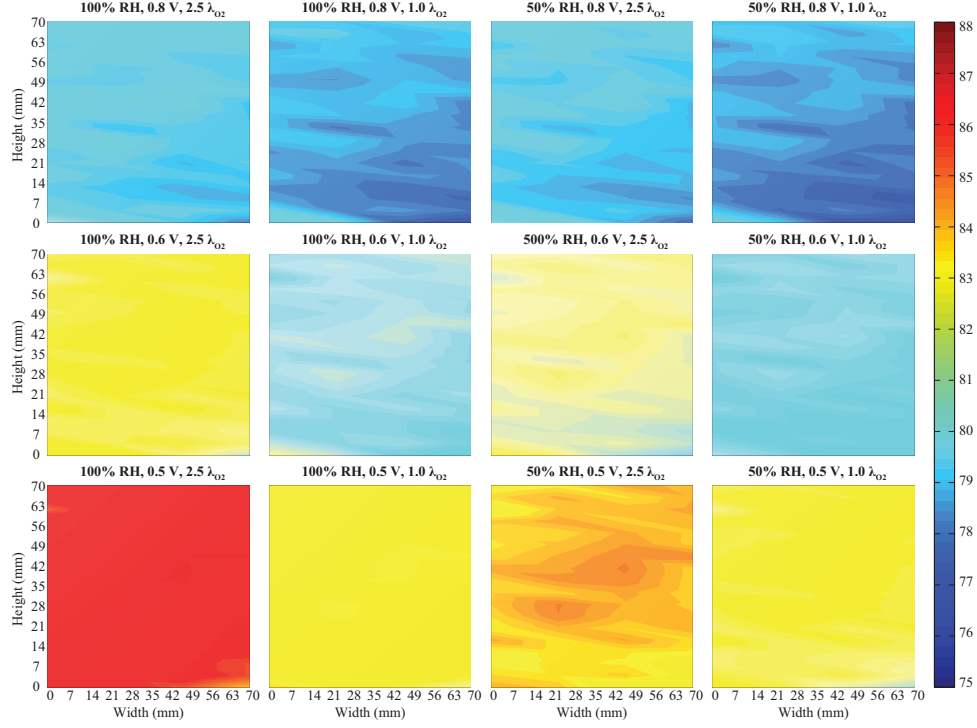


Figure 4: Steady state temperature distributions at constant voltage

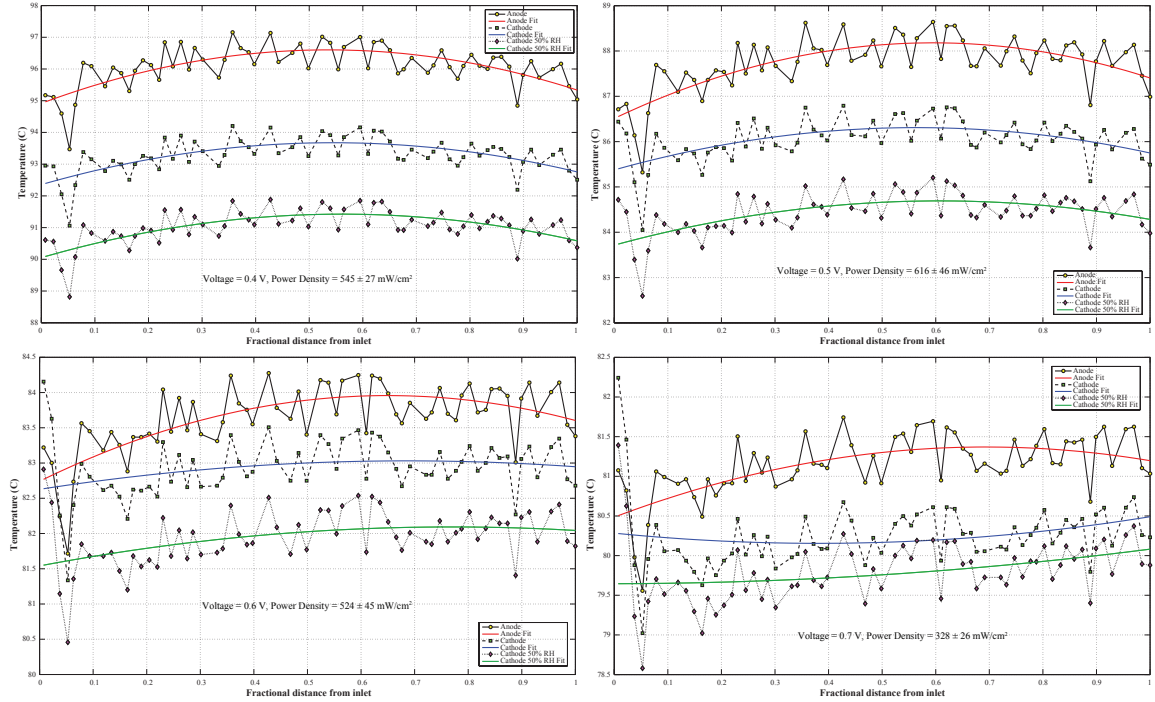


Figure 5: Steady state temperature distributions along the fractional distance from inlet

3.4 Impact of uncertainty on dynamic temperature response

Fuel cell transient temperature behaviour under different air humidity condition for step change in voltage and air stoichiometry is shown in Figure 7. We observe that at 0.5 V for 2.5 air

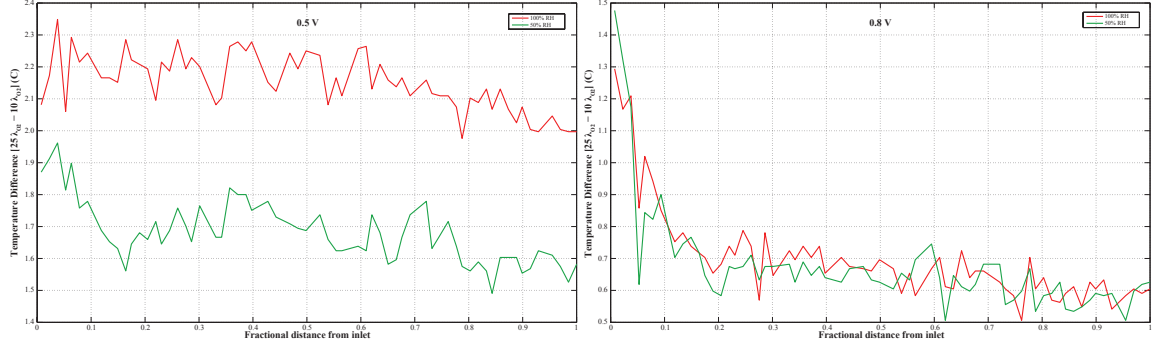


Figure 6: Difference in temperature distributions at 0.5 and 0.8 V

stoichiometry temperature is higher than at the value of $1.0 \lambda_{O_2}$. Low air stoichiometry decreases the rate of water removal which decreases cell performance while at high air stoichiometry the availability of oxygen improves the cell performance which gives rise to increase in reaction temperature. However, the high temperature dries the cell out and we observe the current density response to deteriorate due to loss of catalyst mass as well as its activity, [4]. This deterioration is clearly seen from the step change from 0.5 to 0.4 °C where temperature rises beyond the maximum safe operating temperature. The dynamic response clearly highlights the role of air stoichiometry and its relative humidity as a manipulated variable to control to temperature distribution. Figure 8 presents the snapshots of thermal map as it evolves in time. We observe that the hotspots are 2 to 4 °C higher than the surrounding temperatures. These temperature hotspots are significantly hotter at air stoichiometry 2.5 compared to 1.0. This dynamic data can be used to not only to validate the mathematical model but also to identify exact locations and time to peak of temperature fronts at varying operating conditions.

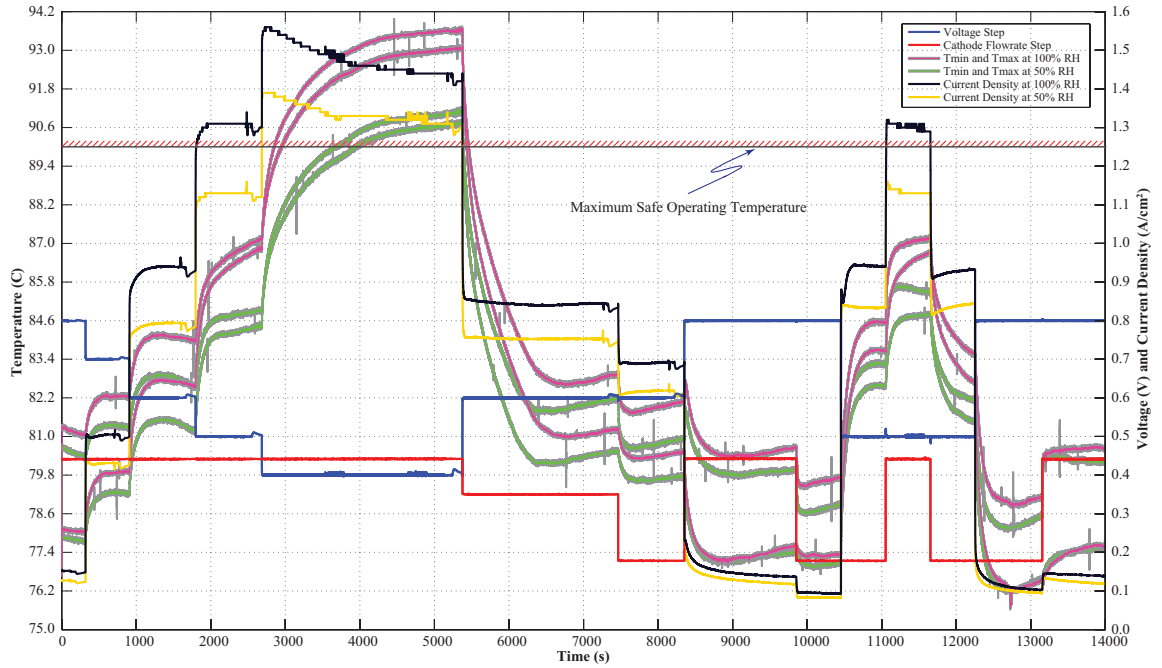


Figure 7: Dynamic evolution for step change in voltage and cathode flowrate

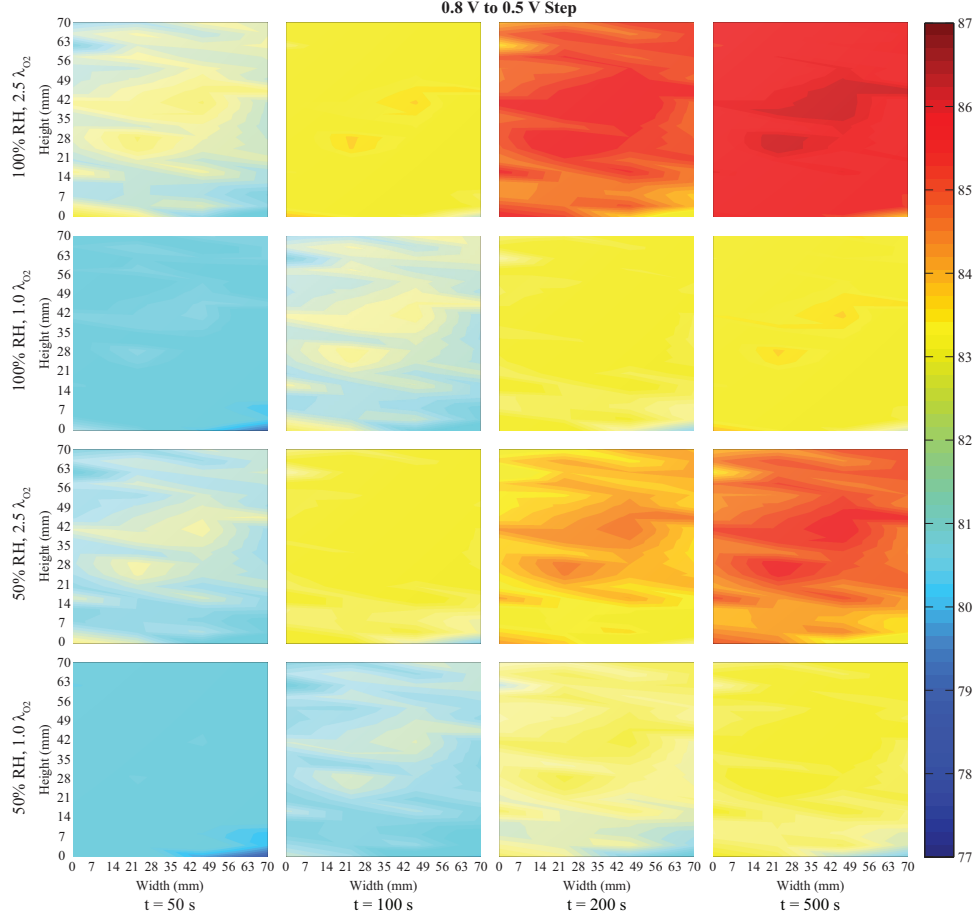


Figure 8: Snapshots of temperature distribution for 0.8 to 0.5 V step

3.5 Impact on internal resistances

Figure 8 shows the impact of uncertainty on internal resistance. We observe that the fuel cell internal resistance increases when the air relative humidity decreased from 100% to 50%. The resistance values when the fuel cell was operated at a 800 mA/cm^2 are: 0.2 cm^2 at 100% RH and 0.4 cm^2 at 50% RH. The increase in resistance at reduced humidity accelerates membrane degradation and can ultimately result in pinhole formation permitting reactant gas crossover. The higher stoichiometry increases humidification level, consequently the performance recovers. Impact of uncertainty has a stronger influence at higher current density when the internal resistance is significantly higher in magnitude. In other work, [12], we measure the internal resistances of the cell and observed that the temperature distributions have a significant effect on the fuel cell performance as it affects internal resistances which ultimately influences membrane conductivity or water transport.

4 Conclusion

Cathode performance is crucial for the operation of a fuel cell because of the slow nature of oxygen reduction reaction. Surmounted by that fact, the results presented in this work clearly show

that the uncertainty in input air quality has significant impact on not only the operation but also material properties of the fuel cell. At low air stoichiometry ability to remove water reduces, which affects the oxygen diffusion to the catalyst sites causing reduction in performance. This impact of uncertainty which adversely affects the electrochemical reaction is critical for the thermal management and durability of the cell. We identified the optimal operating zone using polarization characteristics that depends on the temperature changes within the cell. Based on these results we can say that the air stoichiometry and its humidity are important manipulated variables to control the temperature hotspots and improve the overall operation performance of the fuel cell.

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